Hong Wang（王虹）

Hello! I'm Hong Wang. I did my PhD with Prof. Larry Guth at MIT in 2019.

I'm interested in Fourier analysis and related problems. For example, if we know that the Fourier transform of a function is supported on some curved objects, a sphere, or some "curved" collection of discrete points, what can we say about this function? How to decompose this function into pieces in a meaningful way (which is related to the decoupling theory)? It turns out that such problems are also related to the Falconer distance problem and incidence geometry and I'm interested in these connections.

2019-2021, I was a postdoc member at the IAS ; 2021-2023, I was an assistant professor at UCLA.

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Institut Mittag-Leffler program: Interactions between fractal geometry, harmonic analysis and dynamical systems. September 2-December 11, 2026. Co-organize with Michael Björklund, Tuomas Orponen, Tuomas Sahlsten, and Pablo Shmerkin.

Publications and preprints

0. Streamlined version of the proof of the Kakeya set conjecture in R^3. with L. Guth and J. Zahl.
1. Short proofs of discretized projection theorems, with W. O'Regan and P. Shmerkin.
2. Heilbronn's triangle problem in three dimensions, with D. Maldague and D. Zakharov.
3. Two-ends Furstenberg estimates in the plane , with S. Wu.
4. Volume estimates for union of convex sets, and the Kakeya set conjecture in three dimensions. with J. Zahl.
5. Restriction estimates using decoupling theorems and two-ends Furstenberg inequalities, with S. Wu. (This paper supersedes 16.)
6. Szemeredi-Trotter bound for tubes and applications with C. Demeter, **Ars Inveniendi Analytica** (2025).
7. L^p weighted Fourier restriction estimates with X. Du, J. Li and R. Zhang.
8. The Assouad dimension of Kakeya sets in $\mathbb{R}^3$ with J. Zahl. **Invent. Math** (2025)
9. How much can heavy slices cover? with D. Dabrowski and T. Orponen, **JLMS** (2024).
10. Furstenberg sets estimate in the plane , with K. Ren.
11. Some sharp inequalities of Mizohata--Takeuchi-type , with A. Carbery and M. Iliopoulou, **Revisita** (2024).
12. Dimensions of Furstenberg sets and an extension of Bourgain's projection theorem , with P. Shmerkin, to appear in **APDE.**
13. A restricted projection problem for fractal sets in $\mathbb{R}^n$, with S. Gan and S. Guo, accepted by **Cambridge Journal of Mathematics.**
14. Sticky Kakeya sets and the sticky Kakeya conjecture , with J. Zahl, accepted by **JAMS.**
15. A dichotomy for H\"{o}rmander type oscillatory integral operators, with S. Guo and R. Zhang, **Invent. Math** (2024).
16. An improved restriction estimate in $\mathbb{R}^3$, with S. Wu. (This paper is superseded by 5)
17. Kaufman and Falconer estimates for radial projections and a continuum version of Beck's theorem , with T. Orponen and P. Shmerkin, **GAFA** (2024).
18. On restricted projections to planes in $\mathbb{R}^3$, with S. Gan, S. Guo, L. Guth, T. Harris, D. Maldague, accepted by **Amer. J. M.**
19. On the distance sets spanned by sets of dimension d/2 in $\mathbb{R}^d$ with P. Shmerkin, accepted by **GAFA.**
20. The Bochner-Riesz problem: an old approach revisited, with S. Guo, C. Oh, S. Wu and R. Zhang, **Peking Math Journal** (2024).
21. Improved decoupling for the parabola , with L. Guth and D. Maldague, **JEMS**(2024).
22. An improved result for Falconer's distance set problem in even dimensions, with X. Du, A. Iosevich, Y. Ou, and R. Zhang, **Math. Annalen** (2021).
23. Optimal Analysis of Subset-Selection Based l^p Low-Rank Approximation, with C. Dan, H. Zhang, Y. Zhou, P. Ravikumar, **NeurIPS 2019.**
24. 2D-Defocusing Nonlinear Schrodinger equation with random data on irrational tori with C. Fan, Y. Ou and G. Staffilani, **Stoch. Partial Differ. Equ. Anal. Comput.** (2021)
25. A sharp square function estimate for the cone in $\mathbb{R}3$ with L. Guth and R. Zhang, **Ann. of Math** (2020).
26. Small cap decouplings with C. Demeter and L. Guth, with appendix by Roger Heath-Brown, **GAFA** (2020).
27. Incidence estimates for well spaced tubes with L. Guth and N. Solomon, **GAFA** (2019).
28. Lower bounds for estimates of the Schr\" odinger maximal function with X. Du, J. Kim and R. Zhang, **Math. Res. Let.** (2020)
29. On Falconer's distance set problem in the plane with L. Guth, A. Iosevich and Y. Ou, **Invent. Math**(2019).
30. Weighted restriction estimates and application to Falconer distance set problem with X. Du, L. Guth, Y. Ou, B. Wilson and R. Zhang, **Amer. J. Math**. (2021)
31. A restriction estimate in $\mathbb{R}^ 3$ using brooms. **Duke Math. J.** (2022).
32. A cone restriction estimate using polynomial partitioning with Y. Ou, **JEMS**(2022).
33. On a bilinear Strichartz estimate on irrational tori and some application with C. Fan, G. Staffilani and B. Wilson, **Analysis&PDE**(2018).
34. Refinements of the 2-dimensional Strichartz estimate on the maximum wavepacket with L. Zhang.
35. Decoupling and near-optimal restriction estimates for Cantor sets with I. Laba, **IMRN**(2017).
36. Exposition of Elekes Szabo paper.
37. Bounds of incidences between points and algebraic curves with B. Yang and R. Zhang.